# Response to Reviewers — TCS-D-26-00494

Manuscript: "Distributional Learning of Context-Free Languages under Fixed Finite-Semigroup Typing"

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We thank the Editor and both reviewers for their detailed and constructive reports. Their comments led us to correct the learning algorithm, remove an unused typing component, repair the terminal-rule completeness argument, and substantially reorganize the manuscript.

Unqualified section and statement numbers refer to the revised manuscript; submitted numbers are marked "submitted" or "former". Subheadings such as "Section 7" or "Sections 8–10" follow the reviewers' grouping and refer to the submitted manuscript. We number the Editor's comment E.1 and the reviewers' comments R1.1–R1.37 and R2.1–R2.11; quotations are excerpts with typographical normalization, omissions marked by ellipses, and related remarks grouped by topic. The locator guide below refers to the additions-marked revised manuscript and is checked against the marked PDF. Because the revision substantially reorganizes the paper, the marked PDF displays extensive additions, including passages rewritten for presentation rather than newly claimed results. An unmarked LaTeX source file is supplied separately.

## Overview of changes

**Framework of the revised manuscript.** Reviewer quotations retain the reviewers' original notation. The revised manuscript fixes a surjective finite-semigroup typing $s : \Sigma^+ \to \mathsf{S}$, with the fresh-identity completion $\widehat{s}$ distinguishing empty yields. This is more than a change of terminology. In the submitted Definition 2.3, the relation induced by $h : \Sigma^* \to M$ was tested on all $x, y \in \Sigma^*$, including $\lambda$ and positive words of type $h(\lambda)$. The revised definition uses the positive restriction $s = h|_{\Sigma^+}$ and tests only nonempty factors, while $\widehat{s}$ handles empty yields separately (Lemma 2.2). Thus each submitted fixed-$h$ target class is contained in the corresponding revised fixed-$s$ class; we do not claim the two definitions are equivalent. The title has been changed accordingly. Section 5 now types arbitrary CFG productions directly, including unit, nullable, and mixed right-hand sides, without start-symbol separation. BUNF (Definition 6.1) is retained only for the quantitative bounds in Sections 6–8. The sequential learner remains conservative; no incremental polynomial-data bound is claimed.

**Central corrections.** The revision makes four central corrections. First, it separates the set-driven reconstruction operator $\mathsf{Rec}_s$ from the conservative sequential learner $\mathsf{Learn}_s$, so that the Gold learner now stabilizes syntactically. Second, the unused outer-context type annotations are removed, and the occurrence-indexed hypothesis symbols are quotiented to one substring-indexed symbol $[\![x]\!]$ per observed factor. The resulting batch construction follows Yoshinaka's substring grammar (as also presented by Clark), replacing the window comparison in the unary rules by equality of $s$-types. Third, the terminal-rule completeness gap is repaired: witnesses for typed terminal rules, already implicit in the submitted characteristic sample, are now listed explicitly, and the completeness proof uses unary Rule (U) before lexical Rule (L) when necessary. Fourth, fixed-$s$ substitutability is stated for nonempty internal fragments, while $\lambda$ is distinguished in typed grammars by the fresh identity of $\mathsf{S}^\sharp$.

**Scope and prior work.** The revised Introduction cites the author's 2014 preprint (arXiv:1409.6247v3) as historical background and acknowledges Yoshinaka's (2015) general disjoint-sort framework, which also supplies positive-data identification, polynomial-time hypothesis updates, and polynomial-cardinality characteristic data under the corresponding many-sorted specialization. These general learning guarantees and fixed-window signatures are not presented as independent novelties. The semigroup identities and recognizable-language background of Proposition 3.5 are attributed to Pin's published 1986 monograph; the positive-image/kernel formulation is proved explicitly in the revised manuscript. Yoshinaka (2008, Theorem 1) already provided polynomial encoded-size characteristic-data bounds for fixed-window classes, and Yoshinaka (2015) supplies the qualitative learning framework and polynomial characteristic-data cardinality under the corresponding many-sorted specialization. The revised paper's quantitative contribution instead concerns encoded-length bounds beyond fixed windows in the stated window–group and linear regimes, together with a fixed-typing exponential lower bound despite small ordinary grammar thickness. We also give expressive separations and obstructions. The elementary retyping-closure facts and classical fixed-window criteria are recorded to delimit the framework, rather than claimed as independent new closure or semigroup theorems. The relevant prior work is cited in the manuscript and addressed in the responses below.

**Narrowed claims.** Presentation-order independence and polynomial characteristic-data bounds are claimed only for the set-driven operator $\mathsf{Rec}_s$, the latter only in the stated quantitative regimes; the sequential learner $\mathsf{Learn}_s$ is proved to have Gold convergence and polynomial-time updates, without an incremental polynomial-data guarantee (Section 2.1; R1.1). The submitted Section 7 title is withdrawn accordingly (R2.3). The former

claim that outer-context types are necessary is withdrawn, and the quotient construction needs no separate context-transport rule (R2.2).

**New quantitative results.** Proposition 6.4 gives finite Clark–Eyraud substitutable languages $L_n$ with grammars of size and ordinary thickness $O(n)$: under trivial typing $\mathsf{Rec}_{s_0}$ has polynomial characteristic data, whereas under a fixed two-element semilattice typing every characteristic sample for $\mathsf{Rec}_{s_U}$ contains a word of length $3 \cdot 2^n - 2$. By Proposition 6.4(iv), the same holds for any set-driven operator admitting characteristic samples for both $L_n$ and $L_n \setminus \{w_n^*\}$ in the fixed-typing class. Corollary 6.5 gives $\tau_{s_U}^{\mathrm{typ}}(G) \geq 3 \cdot 2^n - 2$ for every reduced BUNF grammar $G$ of $L_n$, regardless of size, and Lemma 7.6, an elementary semigroup fact recorded for completeness, shows $U_1 \notin \mathbf{LI} \vee \mathbf{Grp}$; hence the algebraic condition of Corollary 7.5 cannot simply be dropped from a uniform characteristic-data bound. Neither necessity for every typing nor any lower bound on the running time or convergence stage of arbitrary Gold learners is claimed (R2.1).

**Reorganization and clarifications.** The manuscript has been substantially shortened (R1.2). Definitions are consolidated in Section 2; the reconstruction operator precedes the target-side refinement used for completeness; Sections 6–8 reorganize the quantitative discussion and answer Reviewer 2's question on the linear theorem (R2.3); the three former boundary sections form Section 9, which also contains Proposition 9.7 suggested by Reviewer 1; normalization proofs are in Appendices A–B; and Appendix C proves Proposition 6.4(i) by a reversible deterministic parser. The closure facts of the submitted Appendix A are condensed in Proposition 3.2 (R1.36). For the normalized linear grammar $G_0$, Theorem 8.3 is made explicit as $\|\mathsf{Wit}(\widetilde{G}_0)\| = O(|G_0|^3 |\mathsf{S}^\sharp|^4)$; the same degree is not claimed relative to an unnormalized input grammar. We also explicitly attribute the known fixed-window closure and nonclosure properties to Yoshinaka (2008, Proposition 1) (Section 3), and clarify the factorization $\mathsf{S}^n = \mathsf{S}E(\mathsf{S})\mathsf{S}$ (Proposition 3.5), and the essential-node width estimate in the proof of Lemma 7.1 (Appendix A.1). Changes to Proposition 9.1 and the sharpness of $\rho \geq 2$ are described in R1.33–R1.35.

**Selected renumbering guide.** Because the manuscript was substantially reorganized, the following table collects the submitted items most frequently referred to in the reports.

| Submitted manuscript | Revised manuscript |
|---|---|
| Definition 2.3 ($\sim_s$) | Definition 2.1 |
| Definition 3.2 (Clark–Eyraud) | unnumbered special case, Section 3.1 |
| Proposition 3.5 | Proposition 3.3 |
| Theorem 3.7 | Theorem 3.4 |
| Example 4.3 | removed |
| Submitted Rules (2), (3) | quotient construction; unary Rule (U) |
| Lemma 5.2(i) | witness set (2), Theorem 5.2 |
| Theorem 5.7 | Theorem 5.3 |
| Corollary 5.8 | Corollary 5.4 |
| Proposition 7.2 | Proposition 8.1 |
| Theorem 7.9 | Theorem 8.3 |
| Theorem 8.6 | Theorem 9.2 (simplified proof) |
| Corollary 8.7 | removed |
| Propositions 10.3–10.5 | Proposition 9.1; criterion after Proposition 3.3 |
| Lemma 9.11 | Lemma 9.6 |
| Appendix A (closure facts) | Proposition 3.2 (adapted) |
| $\mathrm{CCL}_p$, CCL | $\mathrm{CTR}_\rho$, CTR |
| Counter alphabet $d, u, ;$ | $d \mapsto \uparrow$ (increment), $u \mapsto \downarrow$ (decrement), $; \mapsto \#$ (reset) |
| $L_Y$ (classical language) | $L_{\mathrm{Luk}}$ (Łukasiewicz) |
| Props. 9.2, 9.6, 9.9; Lemma 9.8 | brief statements/citations in Section 9.3 |

**Locator guide for the additions-marked revised manuscript.** Each location below is anchored to a named section or statement, with its actual PDF page and a margin-line number checked in the freshly compiled additions-marked manuscript. A section anchor may cover multiple revised paragraphs; the named statement identifies the relevant paragraph.

| Item | Revised-manuscript locator |
|---|---|
| E.1 | Definition 2.1 (marked PDF p. 4, margin l. 157); Sections 4–5 (marked PDF p. 8, margin l. 319) |
| R1.1 | Section 2.1 (marked PDF p. 3, margin l. 118); Section 4.2 (marked PDF p. 9, margin l. 356); Section 5.2 (marked PDF p. 10, margin l. 405) |
| R1.2 | Introduction (marked PDF p. 1, margin l. 19); Appendices A–C (marked PDF p. 22, margin l. 851) |
| R1.3 | Introduction and related work (marked PDF p. 1, margin l. 19) |

| Item | Revised-manuscript locator |
|---|---|
| R1.4 | Introduction (marked PDF p. 1, margin l. 19); Definition 2.1 (marked PDF p. 4, margin l. 157) |
| R1.5 | Section 1.2 (marked PDF p. 2, margin l. 66) |
| R1.6 | Sections 9.3–9.4 (marked PDF p. 20, margin l. 776) |
| R1.7 | Section 4.2 (marked PDF p. 9, margin l. 356) |
| R1.8 | Section 2 (marked PDF p. 3, margin l. 93); Section 4.1 (marked PDF p. 8, margin l. 320) |
| R1.9 | Section 5.1 (marked PDF p. 10, margin l. 371) |
| R1.10 | Introduction (marked PDF p. 1, margin l. 19); Section 3 (marked PDF p. 5, margin l. 185) |
| R1.11 | Lemma 2.2 (marked PDF p. 4, margin l. 159); Section 3.1 (marked PDF p. 6, margin l. 236) |
| R1.12 | Section 3.1 (marked PDF p. 6, margin l. 236); Proposition 3.3 (marked PDF p. 6, margin l. 247); Proposition 3.5 (marked PDF p. 8, margin l. 300) |
| R1.13 | Section 2 (marked PDF p. 3, margin l. 93); Theorem 3.4 (marked PDF p. 7, margin l. 272) |
| R1.14 | Theorem 3.4 (marked PDF p. 7, margin l. 272) |
| R1.15 | Section 2 (marked PDF p. 3, margin l. 93); Section 5.1 (marked PDF p. 10, margin l. 371) |
| R1.16 | Section 2 (marked PDF p. 3, margin l. 93) |
| R1.17 | Section 5.1 (marked PDF p. 10, margin l. 371) |
| R1.18 | Section 4 (marked PDF p. 8, margin l. 319); Sections 5.1–5.2 (marked PDF p. 10, margin l. 371) |
| R1.19 | Proposition 5.1 (marked PDF p. 10, margin l. 387) |
| R1.20 | Section 5.1 (marked PDF p. 10, margin l. 371) |
| R1.21 | Section 5.1 (marked PDF p. 10, margin l. 371) |
| R1.22 | Proposition 5.1 (marked PDF p. 10, margin l. 387) |
| R1.23 | Proposition 5.1 (marked PDF p. 10, margin l. 387) |
| R1.24 | Section 4.2 (marked PDF p. 9, margin l. 356); Section 5.2 (marked PDF p. 10, margin l. 405) |
| R1.25 | Section 4.2 (marked PDF p. 9, margin l. 356); Section 5.2 (marked PDF p. 10, margin l. 405) |
| R1.26 | Section 4.1 (marked PDF p. 8, margin l. 320) |
| R1.27 | Theorem 4.2 (marked PDF p. 9, margin l. 343) |
| R1.28 | Theorem 5.2 (marked PDF p. 10, margin l. 406) |
| R1.29 | Theorem 5.3 (marked PDF p. 11, margin l. 426) |
| R1.30 | Theorem 5.5 (marked PDF p. 11, margin l. 445) |
| R1.31 | Section 8 (marked PDF p. 16, margin l. 625); Appendix B (marked PDF p. 23, margin l. 904) |
| R1.32 | Section 8 (marked PDF p. 16, margin l. 625); Appendices A–B (marked PDF p. 22, margin l. 851) |
| R1.33 | Theorem 9.2 and Section 9.2 (marked PDF p. 20, margin l. 768) |
| R1.34 | Section 9 (marked PDF p. 18, margin l. 707) |
| R1.35 | Proposition 9.1 (marked PDF p. 19, margin l. 733); Sections 9.2–9.3 (marked PDF p. 19, margin l. 760) |
| R1.36 | Lemma 9.6 (marked PDF p. 20, margin l. 795) |
| R1.37 | Proposition 9.7 (marked PDF p. 21, margin l. 810) |
| R2.1 | Introduction (marked PDF p. 1, margin l. 19); Section 3 (marked PDF p. 5, margin l. 185); Proposition 6.4 (marked PDF p. 13, margin l. 508); Lemma 7.6 (marked PDF p. 15, margin l. 609); Sections 8–9.2 (marked PDF p. 16, margin l. 625) |
| R2.2 | Section 4.1 (marked PDF p. 8, margin l. 320); Sections 5.1–5.2 (marked PDF p. 10, margin l. 371) |
| R2.3 | Theorem 8.3 (marked PDF p. 16, margin l. 648); Proposition 8.4 (marked PDF p. 17, margin l. 667) |
| R2.4 | Sections 4–5 (marked PDF p. 8, margin l. 319); Appendices A–B (marked PDF p. 22, margin l. 851) |
| R2.5 | Proposition 3.1 (marked PDF p. 5, margin l. 191) |
| R2.6 | Section 5.1 (marked PDF p. 10, margin l. 371) |
| R2.7 | Section 5.1 (marked PDF p. 10, margin l. 371) |
| R2.8 | Section 5.1 (marked PDF p. 10, margin l. 371); Theorem 5.2 (marked PDF p. 10, margin l. 406) |
| R2.9 | Theorems 5.2–5.3 (marked PDF p. 10, margin l. 406) |
| R2.10 | Corollary 5.4 (marked PDF p. 11, margin l. 431) |
| R2.11 | Proposition 8.1 (marked PDF p. 16, margin l. 629); Lemma 8.2 (marked PDF p. 16, margin l. 636); Appendix B (marked PDF p. 23, margin l. 904) |

## Response to the Editor

**E.1. Editor (quoted excerpt):** "The reviewers agree that the scientific contribution has merit. However, they both point to many issues with the presentation, many of which leak also into the scientific part of the paper and are not merely language revisions. This is especially relevant in the learning part of the paper. ... Specifically, any correctness issues should be carefully considered."

**Response:** We agree. The submitted learner has been replaced by a conservative Gold learner; the terminal-rule gap in the completeness proof has been repaired, and the terminal-rule witnesses are now listed explicitly; the unused outer-context annotations have been removed. Fixed-$s$ substitutability is now stated for nonempty internal fragments, with $\lambda$ handled separately. Section 4 defines the batch operator and sequential learner before Section 5 introduces target-side yield typing for completeness. Polynomial characteristic data are attributed to the set-driven batch operator, while the sequential learner is asserted to have Gold stabilization with polynomial-time updates.

**Revision location:** Definition 2.1; Section 4.2; Section 5.1 and Theorem 5.2; Corollary 5.4; Sections 6–8.

## Reviewer 1

### General assessment and learning algorithm

**R1.1. Comment (quoted excerpt):** "The main result of this submission is an elegant generalization of the results of Clark and Eyraud (2007) on the learning of substitutable context-free languages (SCFLs) and of Yoshinaka (2008) on the learning of k,l-SCFLs. ... The only serious error is in the definition of their learning algorithm, but this is easily repairable by following prior work."

**Response:** We thank the reviewer for identifying the central correctness problem. We agree that the qualitative identification scheme and many of the arguments are standard or follow earlier work. The revised Introduction therefore identifies these guarantees as specializations of Yoshinaka's 2015 framework, while separately crediting the 2008 fixed-window results; we do not present the qualitative learning theorem or polynomial-time hypothesis updates as independent contributions. Recomputing the batch grammar after every datum need not stabilize syntactically. The revision therefore separates a set-driven operator $\mathsf{Rec}_s$ (Section 4.1) from a conservative sequential learner $\mathsf{Learn}_s$ (Section 4.2), using the same conservative rebuild-on-missed-positive-example pattern as Clark–Eyraud's SGL and Yoshinaka's $(k, \ell)$-SGL. Corollary 5.4 proves eventual syntactic stabilization. As noted by Eyraud–Heinz–Yoshinaka (2016, Appendix), now cited in Section 4.2, the correct rebuild may be delayed until the first positive example missed after a characteristic sample has appeared. Hence $\mathsf{Learn}_s$ is claimed to have Gold convergence with polynomial-time updates, but not the incremental polynomial-time-and-data criterion (ibid., Definition 2.5). Our polynomial characteristic-data guarantees are set-based (ibid., Definition 2.8): they concern $\mathsf{Rec}_s$, only in the stated quantitative regimes, not uniformly for all $s$. Section 2.1 now makes this distinction explicit.

**Revision location:** Section 2.1; Section 4.2; Corollary 5.4; Theorem 7.2; Theorem 8.3 and the following clarification.

**R1.2. Comment (quoted excerpt):** "On the other hand, the presentation leaves considerable room for improvement. By addressing the absence of definitions for technical terms, unnecessarily long proofs, inconsistent use of metavariables, and the over-fragmentation of the paper into too many sections, the authors could substantially reduce the burden on the reader, and I recommend a major revision along these lines. ... I suspect that this material could be presented in less than half the current length."

**Response:** We have reduced the paper from 40 to 26 pages (including the appendices), while maintaining the central proofs and the technical definitions requested by the reviewers. The revised sections consolidate the reconstruction and convergence arguments, the polynomial-data bounds, and the expressive examples. Binary–unit–nullable normal form (BUNF) is defined in Definition 6.1 only as a device for the quantitative estimates. The qualitative witness construction and completeness proof of Section 5 work directly for arbitrary CFG right-hand sides. The quantitative normalization and reversible stack-parser arguments are in Appendices A–C. We have also addressed the reviewer's four presentation concerns: technical terminology is defined at first use or in Section 2; proof-local detail was shortened or moved to Appendices A–C; word metavariables in the Section 9 concatenation identity and quotient argument no longer reuse the typing symbols $s, t$; and the former boundary sections have been consolidated into Section 9. Sections 6–8 remain separate because they concern distinct quantitative regimes (general typings, window–group typings, and linear targets). We acknowledge that the revised paper is shorter but not half the length of the submitted version; it also incorporates the substantive corrections and new results requested by the reviewers.

**Revision location:** Sections 1–2; Sections 4–9; Appendices A–C.

### Section 1

**R1.3. Comment (quoted excerpt):** "Stating the paper's place within the field and the significance of the language families considered would make the paper more compelling. Conversely, the current text is too heavily weighted toward technical content and could be made more concise."

**Response:** Agreed. Section 1.2 now begins with a concise account of the fixed-$s$ batch reconstruction operator and conservative Gold learner as the learning foundation, expressly acknowledging that the learning and polynomial-time guarantees specialize Yoshinaka's general many-sorted signature framework. Rather than listing those existing learning guarantees as separate novel contributions, the Introduction emphasizes two main contributions: (1) quantitative characteristic-data results, including representation sensitivity and the window–finite-group and linear bounds; and (2) structural and expressive results, where the elementary scope-setting facts (retyping closure and the classical fixed-window criterion) are explicitly distinguished from the expressive examples beyond fixed windows, counter/Dyck obstructions, and comparison with Clark's congruential family. The fixed morphism is presented as a finite typing bias (R2.1). The historical account distinguishes the author's

2014 preprint from Yoshinaka's subsequent general many-sorted signature learning theorem and identifies which results are particular to fixed finite-semigroup typing (see also R2.1).

**Revision location:** Section 1.

**R1.4. Comment (quoted excerpt):** "The notations $\mathcal{D}_L$ and $\mathcal{C}_h^{\mathrm{cf}}$ are not standard and require definitions. In particular, the phrasing 'the general context-free class $\mathcal{C}_h^{\mathrm{cf}}$' is liable to be misunderstood."

**Response:** Agreed. $\mathcal{D}_L(x)$ is defined at its first occurrence in the Introduction and again formally in the preliminaries. The notation $\mathcal{C}_s^{\mathrm{cf}}$ is now defined at its first use in the Introduction as the class of context-free $\sim_s$-substitutable languages. The unrestricted fixed-typing class $\mathsf{RS}_s$ is then introduced before $\mathsf{RS} = \bigcup_s \mathsf{RS}_s$. Section 3 develops these classes and $\mathcal{C}_s^{\mathrm{lin}}$ formally. The phrase "general context-free class" has been removed.

**Revision location:** Introduction; Section 2.2 and the opening of Section 3.

**R1.5. Comment (quoted excerpt):** "The contributions are enumerated as (1)-(6), but it is unnecessary to count technical details as separate contributions. Moreover, at this point the text uses many terms whose technical meaning has not yet been made clear, and readers may not be able to follow them."

**Response:** Agreed. Instead of the submitted six-item list, Section 1.2 now gives a short unnumbered paragraph on the qualitative learning construction and its relation to Yoshinaka's general framework, followed by only the two contributions described in R1.3; proof-local devices are not counted. The quantitative item summarizes polynomial upper bounds and exponential lower bounds on characteristic data. Although it mentions *typed thickness* and *binary–unit–nullable normal form*, both are formally defined at the start of the quantitative material in Section 6. The structural item identifies the fixed-window criterion as classical, with *locally trivial* defined in Section 2.2. The Clark–Eyraud (2007, Section 6.1) example $L = \{a, aa\}$, recalled after the fixed-$s$ definition in Section 1.1, shows that trivial typing fails (two factors share a context but have different distributions); parity typing then distinguishes them. It illustrates the definition and is not a separation from fixed-window classes.

**Revision location:** Section 1.1 (minimal example after the definition of fixed-$s$ substitutability); Section 1.2.

**R1.6. Comment (quoted excerpt):** " 'Yoshinaka's classical language' is often called 'the Łukasiewicz language' in the literature."

**Response:** Agreed. We use "Łukasiewicz language," add the Autebert–Berstel–Boasson (1997) handbook reference, and state its relation to the one-bracket Dyck language concisely. We also now cite Yoshinaka (2008, Section 3) for the prior observation that this language is outside every fixed $(k, \ell)$ class, and distinguish our stronger conclusion that it lies outside the finite-typing union $\mathsf{RS}$.

**Revision location:** Section 9.3.

**Section 2**

**R1.7. Comment (quoted excerpt):** "I doubt the claim that 'the learners considered in this paper produce their output independently of the order in which positive examples are observed.' It appears that the learning algorithm is being conflated with the grammar-construction algorithm that it uses as a subroutine. Indeed, the 'learning algorithm' in Section 5 does not converge, and hence does not satisfy Gold's criterion. Even if the learning algorithm were corrected to follow Clark & Eyraud, the output would still differ, when learning, e.g., the language $a^n c b^n$, between the case where the characteristic sample itself is received first and the case where the first positive example received is very long."

**Response:** We withdrew the claim that the sequential learner is order-independent. It is the batch reconstruction operator $\mathsf{Rec}_s$, not the conservative learner $\mathsf{Learn}_s$, that is set-driven. Section 4.2 restores the explicit argument: for any infinite target, take $w$ longer than every word in the sample at the final rebuild on a convergent text. The limiting grammar lacks $[\![w]\!]$, but prepending $w$ yields another text whose rebuilt and limiting grammar contains $[\![w]\!]$. Both limit hypotheses generate the same language. See also Clark (2013, Section 4, Example 1).

**Revision location:** Section 2.1; Section 4.2; Corollary 5.4.

**R1.8. Comment (quoted excerpt):** "This section should thoroughly define the necessary notation and terminology, including basic notions such as (linear) context-free grammars - some terms appear to be defined ad hoc in footnotes. Those terms include 'type', 'reconstruction', 'realize', 'reachable', 'observation', 'control', 'local configuration', etc. If these words carry no mathematical definition required to read the paper and were introduced only to aid intuition, then in the present form they tend to confuse the reader, and in some cases it would be better not to name them at all. A flood of excessive terminology can itself burden the reader and impede smooth understanding."

**Response:** Agreed. Section 2 now consolidates the grammar and learning notions needed later, including linear CFGs, reachable/productive/reduced grammars, the positive-data model, set-driven reconstruction, characteristic samples, distributions, and the fixed-$s$ classes. Proof-local labels without mathematical content have been removed or are introduced only at their point of use (e.g., the former *finite typed reconstruction basis*; R1.9).

Section 4.1 uses the standard substring-indexed factor nonterminals $[\![x]\!]$, defined at first use. The revision also defines $|w|_a$ in Section 2, spells out SGL (*Substitution Graph Learner*) and $(k,\ell)$-SGL ($(k,\ell)$-*Substitutable Grammar Learner*) at their point of use, specifies $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$ at its first appearance, and removes the unused class notation $[x]_{\equiv_L}$. As for the listed terms: *reachable* and yield *type* are defined in the preliminaries and typing setup, respectively; *reconstruction* refers to the defined batch operator $\mathsf{Rec}_s$; *observation* is not a separate technical notion, Section 4.1 speaking only of observed factors and their hypothesis symbols; *realize* and *local configuration* have been removed; and *control* remains only in standard uses such as control sets and pushdown finite control.

**Revision location:** Section 2; Section 4.1; Section 5.1; Appendix A.2.

**R1.9. Comment (quoted excerpt):** "The term 'finite typed reconstruction basis', a combination of undefined words, is clearly defined here; however, if it denotes an atomic concept for which none of its four constituent words is individually defined, a shorter term would be easier to understand."

**Response:** Agreed. Rather than rename this proof-local object, we removed both the term and the basis itself. The revised completeness argument uses only the yield-typed target grammar and the finite witness set $\mathsf{Wit}(\widetilde{G})$ that is actually needed for exact reconstruction.

**Revision location:** Section 5.1.

### Section 3

**R1.10. Comment (quoted excerpt):** "The paper emphasizes (somewhat excessively and awkwardly) that h is fixed. To match this, RS_h should be defined before RS. The family RS is not shown to be learnable from positive examples (and indeed it is not learnable); what is learnable is RS_h."

**Response:** Agreed. The Introduction now explicitly defines $\mathsf{RS}_s$ before $\mathsf{RS} := \bigcup_s \mathsf{RS}_s$, and the Gold-identification theorem covers every *nonempty* language in $\mathcal{C}_s^{\mathrm{cf}} = \mathsf{RS}_s \cap \mathrm{CFL}$ for fixed $s$, not the whole class $\mathsf{RS}_s$; no learnability claim is made for the varying-$s$ union $\mathsf{RS}$, which is used only for expressive comparison. The revision also states that neither $\mathsf{RS}$ nor $\mathsf{RS} \cap \mathrm{CFL}$ is identifiable from positive data on a nonempty alphabet, since these classes contain all finite languages and $\Sigma^*$.

**Revision location:** Abstract; Introduction; Section 2.2 and the opening of Section 3.

**R1.11. Comment (quoted excerpt):** "Definition 3.2 requires care. On this point, the way Clark & Eyraud (2007) state their definition is somewhat careless; however, given that in Section 6.1 of that paper they state that $a^+$ is substitutable, and that their grammar-construction algorithm excludes the empty string, it is reasonable to interpret the intent of their definition as Yoshinaka's (2008) 0,0-substitutability. . . . "

**Response:** We agree with the reviewer's interpretation of Clark–Eyraud (2007): with nonempty internal factors, their intended case coincides with Yoshinaka's $(0,0)$ case (Section 3.1). As suggested, we cite Clark (2013), distinguish Clark–Eyraud's Definition 5 (general strings) from this nonempty-fragment convention, restrict fixed-$s$ substitutability to nonempty internal fragments, and give $\lambda$ a separate fresh-identity type in the general CFG refinement, including nullable non-start nonterminals. New Lemma 2.2 proves that freely adjoining a fresh identity gives an equivalent all-fragment typing formulation, without identifying $\lambda$ with nonempty words of a pre-existing identity type of the positive-image semigroup. The hypothesis reconstruction is unchanged; its target-side completeness proof now treats an arbitrary rule $A \to u_0 B_1 u_1 \cdots B_r u_r$ directly, with typed child yields and empty yields distinguished by the fresh-identity extension $\widehat{s}$. BUNF (Definition 6.1) is used only for the quantitative estimates. Section 3.1 explains the issue for $L = a^+$: $(a,\lambda)$ belongs to both $\mathcal{D}_L(\lambda)$ and $\mathcal{D}_L(a)$, whereas $(\lambda,\lambda)$ belongs only to $\mathcal{D}_L(a)$, so comparing the empty factor would exclude $a^+$. The center-marked language $L_{\mathrm{c}} = \{a^n c b^n : n \geq 0\}$ is used consistently as an illustration; no argument concerning it compares an empty internal factor. The $(k,\ell)$ definition now uses $u_1, u_2, v_1, v_2$ for outer contexts, $p, q$ for the fixed windows, and $r_1, r_2$ for the variable interiors of the compared fragments $pr_1q$ and $pr_2q$.

**Revision location:** Definition 2.1; Lemma 2.2; the Clark–Eyraud special case in Section 3.1; the fixed-window definition in Section 3.1.

**R1.12. Comment (quoted excerpt):** "Proposition 3.5. $\sqcup$ and $\mathrm{pref}_k$ are undefined. . . . The proof is very long. If the multiplication on $T$ (as a monoid) were given explicitly in advance, then one line would suffice to explain that $\tau$ is a monoid homomorphism. Rephrasing in terms of $h_{k,\ell}$ or $M_{k,\ell}$ is needlessly cumbersome."

**Response:** We agree that the fixed-window argument was too long and used unnecessary notation. The symbol $\sqcup$ has been removed, and prefix and suffix notation, including the short-word convention, is defined before use. Proposition 3.3 (submitted Proposition 3.5) now defines fixed windows using a quotient finite semigroup $W_{k,\ell}$ of $\Sigma^+$, rather than a separate four-case multiplication table. The short/long coding is a two-sided congruence, giving an associative quotient product. The same positive-word typing $s_{k,\ell}$ is used in Proposition 3.5, Lemma 7.1, Theorem 7.2, and Lemma 7.4. Short words, on which $(k,\ell)$-substitutability imposes no comparison, remain individually typed. The fixed-window counterexample criterion following Proposition 3.3 summarizes the separation argument used later, and the classical locally trivial-semigroup criterion, the window–finite-group

factorization, and the resulting thick-data bound are stated separately in Proposition 3.5, Lemma 7.4, and Corollary 7.5.

**Revision location:** Proposition 3.3 and the fixed-window counterexample criterion; Proposition 3.5; Lemma 7.4; Corollary 7.5.

**R1.13. Comment (quoted excerpt):** "Rather than saying that $h$ is given as data (the meaning is not crystal clear), it would be cleaner to state early in the paper something like 'fix an efficiently computable $h : \Sigma^* \to M$'. Define 'linear context-free'. ... 'Explicit finite monoid homomorphism' is not defined. ... Technically unnecessary words should be removed, especially in the statements of theorems."

**Response:** Agreed. We fix a surjective finite-semigroup morphism $s : \Sigma^+ \to \mathsf{S}$ by its multiplication table and letter images, and compute its values on nonempty words in linear time. Linear context-free grammars are defined in the preliminaries and the main theorem assumes a fixed efficiently computable $s$.

**Revision location:** Sections 2–3.2.

**R1.14. Comment (quoted excerpt):** "I do not understand 'The proof for fixed $h$'. Do you mean 'The proof for Theorem 3.7'?"

**Response:** Agreed. This stray phrase has been deleted; the revised text states the relevant theorem and its proof directly.

**Revision location:** Section 3.2.

**Section 4**

**R1.15. Comment (quoted excerpt):** "Definition 4.2. 'productive' and 'reachable' are undefined. The phrasing 'This section proceeds in the order typed local control → finite typed basis.' is too casual."

**Response:** Agreed. These notions are defined in the preliminaries, and the typed grammar is defined as the reduced part of the full refinement, so repeated qualifications are unnecessary. The quoted informal sentence has been removed.

**Revision location:** Section 2; Section 5.1.

**R1.16. Comment (quoted excerpt):** "Footnote 2: 'productive' is already used earlier. Define it in Section 2 rather than in a footnote."

**Response:** Agreed. Reachable and productive are now defined in Section 2 before the target refinement is introduced, and the former explanatory footnote has been removed.

**Revision location:** Section 2; Section 5.1.

**R1.17. Comment (quoted excerpt):** "The text says 'the set of reachable and productive typed nonterminals'; is there any nonterminal of $\widetilde{G}$ that fails this condition? This may be intended as a helpful reminder or emphasis, but it confuses the reader instead."

**Response:** The reviewer is correct: every symbol retained in $\widetilde{G}$ is reachable and productive because $\widetilde{G}$ is defined as the reduced part of the full yield-typed refinement. The repeated qualification has therefore been removed.

**Revision location:** Section 5.1.

**R1.18. Comment (quoted excerpt):** "The first paragraph states no reason why 'different occurrences must be treated as different typed copies' at all. Expressions making clear 'for what purpose and for what reason' are needed. Example 4.3 is labeled '(why typing is needed)', yet why it is needed is not stated. ... If the authors wish to explain the reason, the development would be smoother if they first presented the learning algorithm and then, within its completeness proof, stated: 'here we assume the target grammar is type-refined, and we show that the learner's hypothesis grammar can mimic it.' "

**Response:** We have adopted the reviewer's suggested order of presentation. Section 4 first defines the batch reconstruction operator $\mathsf{Rec}_s$ and the conservative sequential learner $\mathsf{Learn}_s$, with no target refinement assumed. Section 5 then introduces the yield-typed target grammar specifically for the completeness proof and states explicitly that the purpose is to show that the already-defined hypothesis grammar can mimic typed target derivations. Outer-context typing has been removed (R2.2). The remaining yield typing supports the unary Rule (U) step of that simulation for arbitrary right-hand sides. For a typed production $\rho : X \to u_0 Y_1 u_1 \cdots Y_r u_r$, its canonical yield $t_\rho = u_0 \omega(Y_1) u_1 \cdots \omega(Y_r) u_r$ satisfies $\widehat{s}(\omega(X)) = \widehat{s}(t_\rho)$. When both yields are nonempty, their common $s$-type and an observed shared context license Rule (U); Rule (B) and Rule (L) then handle the right-hand-side components. Empty-type yields contribute only $\lambda$ and are treated separately.

**Revision location:** Section 4; Section 5.1 and Theorem 5.2.

**R1.19. Comment (quoted excerpt):** "Lemma 4.4. It is confusing that an expression concerning derivations in $G$ appears immediately after 'In the full typed refinement $\widetilde{G}^{\mathrm{full}}$,'. It is unnecessary or should be moved to the end of the sentence."

**Response:** The former Lemma 4.4 has been removed in the reorganization. Its needed content is now folded into Proposition 5.1, which states language preservation and the yield invariant directly and keeps derivations of the original grammar and of the typed refinement syntactically separate.

**Revision location:** Proposition 5.1.

**R1.20. Comment (quoted excerpt):** "Lemma 4.8. 'If $X$ is reachable and productive in $\widetilde{G}$,' This likewise confuses the reader."

**Response:** Resolved by the reduced-refinement convention of R1.17: the former Lemma 4.8 is deleted together with this redundant hypothesis.

**Revision location:** Section 5.1.

**R1.21. Comment (quoted excerpt):** "The opening paragraph (the definitions of $\omega$ and $\chi$) would connect better if placed immediately before the definition of CS."

"Since the term 'characteristic sample' is defined in Def. 2.1, avoid writing 'Define the characteristic sample by' here unless it is immediately clear that this set $CS(G)$ satisfies Def. 2.1."

"The term 'realizable' is defined here but used earlier, whereas the term 'realized', used in many places, is undefined."

"The phrase 'canonical observation word' could be defined when $CS(\widetilde{G})$ is defined."

**Response:** Agreed. The canonical yield $\omega(X)$ is defined after the trimmed refinement, before first use, and the canonical context $\chi(X)$ immediately before the finite witness set. $\mathsf{Wit}(\widetilde{G})$ is called a witness set until exact reconstruction has been proved, and only then identified as a characteristic sample. The unused basis, $NT(\cdot)$, Rule$(\cdot)$, and the realized/realizable and exhibited terminology have been removed.

**Revision location:** Section 5.1.

**R1.22. Comment (quoted excerpt):** "Lemma 4.6. This is far too trivial a corollary. Either label it as a Corollary or delete it."

"Lemma 4.7. Given what the reader has understood so far, this is trivial and needs no proof. . . . "

"Theorem 4.9. I cannot understand what this theorem shows. . . . "

"Theorem 4.12. The first sentence should be removed. The second and third sentences are not understandable owing to the undefined term 'exhibited'."

**Response:** The former Lemmas 4.6–4.7 and Theorems 4.9 and 4.12 have been removed. The typed-refinement development now consists of a general-CFG type lifting invariant (Proposition 5.1) and a general right-hand-side witness construction. The BUNF definition occurs only in the quantitative Section 6.

**Revision location:** Section 5.1.

**R1.23. Comment (quoted excerpt):** "$\alpha$ is a string, not a symbol."

"Said 'In particular, if $X \to YZ$,' but the antecedent of the 'if' is unrelated to its consequent."

"Lemma 4.5. 'For (iii), every derivation in $G$ erases typed annotations' I cannot understand this. Do you mean 'For (iii), every derivation in $G$ with typed annotations erased gives . . . '?"

"Is $NT(\widetilde{G})$ not the same as $\widetilde{V}$ on page 9?"

"Definition 4.11. Is Rule$(\widetilde{G})$ different from $\widetilde{P}$ on page 9? . . . "

**Response:** Agreed. These sentences disappear in the rewrite, and $NT(\widetilde{G})$ and Rule$(\widetilde{G})$ are replaced by $\widetilde{V}$ and $\widetilde{P}$. The lifting statement is now Proposition 5.1 (R1.19); its proof says that erasing type annotations maps a typed derivation to a derivation of the original grammar, which addresses the question on former Lemma 4.5(iii).

**Revision location:** Section 5.1.

### Section 5

**R1.24. Comment (quoted excerpt):** "The learning algorithm as defined here does not converge. This is the fatal error of the paper."

**Response:** We agree. The submitted learning algorithm did not converge, and this was a central correctness error. As detailed in R1.1, we replaced it with a conservative sequential learner and proved syntactic stabilization in Corollary 5.4.

**Revision location:** Section 4.2; Corollary 5.4.

**R1.25. Comment (quoted excerpt):** "Corollary 5.8. This cannot be accepted as a proof. By the definition in Section 5.1, the learner's hypothesis grammar keeps growing as it receives more positive examples."

**Response:** See R1.1 and R1.24. We changed the learner rather than only the proof; Corollary 5.4 is the resulting stabilization statement.

**Revision location:** Section 4.2; Corollary 5.4.

**R1.26. Comment (quoted excerpt):** "Remark 5.1. I cannot grasp the point. What does 'soundness of a rule' mean? ... If one sentence is to be added, something like 'the grammar is designed so that the language derived by each nonterminal $[x : u, v]$ should be the set of strings substitutable with $x$' might aid intuition. Stating a mathematical property concisely helps the reader more than deploying adjectives that carry no mathematical definition."

"Lemmas 5.3 and 5.4 merely repeat definitions and should be deleted."

**Response:** The remark and definition-restating lemmas are deleted. The hypothesis now has one symbol $[\![x]\!]$ per observed nonempty factor rather than one per occurrence context (R2.2), and, in the spirit of the reviewer's suggestion, the mathematical property is stated directly: whenever $[\![x]\!] \Rightarrow^* w$, we have $x \equiv_L w$ and $s(w) = s(x)$ (R1.27). We do not claim that a finite-sample nonterminal generates the entire syntactic-congruence class.

**Revision location:** Section 4 (Rules (B), (U), (L), (S) and the soundness invariant).

**R1.27. Comment (quoted excerpt):** "Lemma 5.2. In the second sentence, the first claim is trivial and the second is not understandable. What is the corresponding local configuration? Its correspondence with the part following 'More precisely' is unclear."

**Response:** Agreed. The former Lemma 5.2 and the undefined "corresponding local configuration" terminology have been removed. Their role is replaced in the proof of Theorem 4.2 by the simpler invariant

$$[\![x]\!] \Rightarrow^* w \implies x \equiv_L w \text{ and } s(w) = s(x),$$

proved by induction on derivation-tree height. The unary case uses one shared sample context and fixed-$s$ substitutability; the branching case uses that $\equiv_L$ is a monoid congruence. Start rules then give hypothesis soundness.

**Revision location:** Theorem 4.2 and its displayed soundness invariant.

**R1.28. Comment (quoted excerpt):** "Page 14, line 1: 'Since $G$ is in SSBNF' $\to \widetilde{G}$?"

**Response:** Corrected during the rewrite; the relevant grammar is named explicitly throughout the completeness proof.

**Revision location:** Theorem 5.2.

**R1.29. Comment (quoted excerpt):** "Theorem 5.7. Make the statement mathematical. What do you mean by 'a finite sample exposes the finite typed reconstruction basis'? A sample is a string set and the FTRB is a tuple."

**Response:** Theorem 5.3 now states the exact implication $\mathsf{Wit}(\widetilde{G}) \subseteq K \subseteq L \Rightarrow L(\mathsf{Rec}_s(K)) = L$ with all assumptions in the theorem statement.

**Revision location:** Theorem 5.3.

### Section 6

**R1.30. Comment (quoted excerpt):** "If the paper adopts the stance of Remark 6.4, then the fact that the grammar construction can be carried out in polynomial time is almost trivial, so the content of this section could be presented in a single paragraph."

**Response:** The former standalone complexity section is now the short Section 5.3, essentially one counting argument (Theorem 5.5): with $n_K = \sum_{w \in K}(|w| + 1)$, which also counts $\lambda$, there are $O(n_K^2)$ factor nonterminals and $O(n_K^3)$ branching and unary candidates, giving an explicit $O(n_K^4)$ worst-case construction time. The update-time consequence is stated in prose rather than as a separate corollary.

**Revision location:** Section 5.3.

### Section 7

**R1.31. Comment (quoted excerpt):** "This is a special case of the arguments in the preceding sections and could be much shorter ... Too long explanation may blur the focus — what the reader should understand, and what differs from the preceding sections."

"Lemma 7.3. 'strict linear derivation' is not defined."

**Response:** We substantially shortened the material corresponding to submitted Section 7, now Section 8. The main text retains only Proposition 8.1 and Lemma 8.2, which provide the single-spine form and short-yield bound

used by Theorem 8.3; the short proof of Lemma 8.2 follows its statement in Section 8, while the normalization proof, now written with uniform spine chains, is in Appendix B. The undefined term *strict linear derivation* has been removed.

**Revision location:** Proposition 8.1 and the surrounding discussion; Lemma 8.2; Appendix B.

**R1.32. Comment (quoted excerpt):** "Proposition 7.2 does not need two full pages on the proof. I do not know whether exactly the same restriction has been defined before, but it is a combination of existing restrictions that are known to be inessential, and most readers will accept Proposition 7.2 as trivial or as folklore."

"Lemmas 7.5 and 7.6 are too trivial."

**Response:** Agreed. The former two-page normalization proof is replaced by the arrangement described in R1.31 (Proposition 8.1 with its proof in Appendix B, and Lemma 8.2 with a short proof); the former Lemmas 7.5 and 7.6 are absorbed into those arguments or deleted. In line with the folklore remark, Appendix A.2 cites standard terminal-isolation and binarization references (Autebert–Berstel–Boasson, 1997; Hopcroft–Motwani–Ullman, 2007), while Appendix B verifies the one-wrapper-child spine structure with polynomial grammar size and construction time.

**Revision location:** Proposition 8.1 and the surrounding discussion; Lemma 8.2; Theorem 8.3 and the following clarification; Appendices A.2 and B.

**Sections 8–10**

**R1.33. Comment (quoted excerpt):** "The choice of the alphabet $\{d, u, ;\}$ is unfortunate. The semicolon in particular is very hard to read within the running text, and $u$ is a source of confusion because it almost always denotes a string elsewhere in the paper. . . . "

"The proof of Theorem 8.6 should be simplified. Introducing complex formulas and lavishing Greek letters in an attempt to shorten the witness is not the essence of the theorem and only burdens the reader. . . . "

**Response:** Agreed. We renamed $d \mapsto \uparrow$ (increment), $u \mapsto \downarrow$ (decrement), $; \mapsto \#$ (reset), and $p$ as $\rho$, and adopted the reviewer's suggested witness in its essential form, for which we are grateful: Theorem 9.2 now uses $(\uparrow\downarrow)^k \uparrow^{\rho-1} \#^\ell$ and $(\uparrow\downarrow)^k \uparrow^\rho \#^\ell$, which share the empty context and the same fixed-window boundaries, while a preceding $\uparrow$ distinguishes their distributions. The criterion following Proposition 3.3 avoids an expanded six-variable argument. A footnote justifies the sharpness of $\rho \geq 2$ by showing that $\mathrm{CTR}_1$ is $(1, 1)$-substitutable.

**Revision location:** Section 9.2.

**R1.34. Comment (quoted excerpt):** "There is no need to devote an entire subsection to this material. It is unclear what Corollary 8.7 establishes. . . . Proposition 3.5 and Theorem 8.6 are meaningful; but folding them into Corollary 8.7 actually obscures the paper's important contributions."

" 'We note that a block of a word in CCL refers to each of the subwords delimited by the separator symbol ;.' This is a new definition, not a note. In fact, the proof of Theorem 8.6 uses 'block' of a CCL string in an entirely different sense."

"Sections 8-10 all concern the richness of the class of the learning target languages; consolidating them into a single section and keeping the explanations and proofs concise would help the reader grasp the overall picture."

**Response:** We agree that a union of separately learnable families need not itself be learnable. The former Corollary 8.7 has therefore been deleted: RS is used only for expressive comparison, whereas learning is asserted for fixed-$s$ context-free slices, and submitted Proposition 3.5 and Theorem 8.6 remain separate statements (Proposition 3.3 and Theorem 9.2). The former three boundary sections are consolidated in Section 9, with concise proofs. The overloaded *block* terminology is replaced by explicit factors and height-based descriptions, and the witnessing typing $s_\rho = h_\rho|_{\Sigma_c^+}$ of $\mathrm{CTR}_\rho$ is the positive restriction of the DFA transition-monoid morphism $h_\rho$ in Section 9.2, where Yoshinaka's earlier regular separation (2008, Proposition 1) is credited. The submitted Section 10 example $\Delta^* = (\{a^n b^n : n \geq 0\})^*$ is now the shorter Proposition 9.1 (see R1.35). The Kleene-star nonclosure example based on $\{a^n b a^n : n \geq 0\}$ and the appendix on $\mathrm{CCL}_1$ have been deleted.

**Revision location:** Section 9.

**R1.35. Comment (quoted excerpt):** "Definition 9.1. Define the symbol #, and make its use consistent in Section 9."

"Proposition 9.2 is trivial and requires no proof. Proposition 9.6 is also a well-known fact, as already stated in Ginsburg & Greibach, Deterministic context-free languages, Information and Control 9 (1966), 620-648."

"Lemma 9.8. The relationship between the Łukasiewicz language and the Dyck language is also well known."

"The proof of Proposition 10.3 may also be omitted or greatly shortened. In the proof of Proposition 10.5, consider aligning the symbols with those of Definition 3.3."

**Response:** The reset symbol # is defined once and used consistently. The deterministic context-freeness of the uncapped counter CTR (former Proposition 9.2) is stated briefly via a deterministic pushdown counter with reset. For $D_1$ (former Proposition 9.6) and the Łukasiewicz language, we cite Ginsburg–Greibach (1966); the identity $L_{\mathrm{Luk}} = D_1 b$ (former Lemma 9.8) is cited to Autebert–Berstel–Boasson (1997) (see also R1.6). The proof of submitted Proposition 10.3 is condensed in Proposition 9.1: deterministic context-freeness is a one-sentence pushdown argument, and substitutability is proved directly from admissible entry-height sets $\mathcal{Q}(x)$ and a footnoted concatenation identity, separating factors that contain $ba$ from factors in $a^*b^*$. The former Proposition 10.5 proof is replaced by the criterion following Proposition 3.3, and the capped-counter regularity argument is now one elementary DFA paragraph.

**Revision location:** Section 9.

**R1.36. Comment (quoted excerpt):** "Lemma 9.11 is a fundamental property of $\sim_h$ substitutability; since the basic mathematical properties of $\sim_h$ substitutability are collected in an appendix, placing it there is one option."

**Response:** The submitted appendix of basic properties is no longer retained: its intersection and inverse-homomorphism facts are condensed in Proposition 3.2, with erased factors distinguished under the nonempty-fragment convention. The remaining right-quotient property is the two-line Lemma 9.6, used immediately afterwards, so we keep it in place rather than create an appendix for one fact.

**Revision location:** Lemma 9.6.

#### Section 11

**R1.37. Comment (quoted excerpt):** "It is worth mentioning that $\mathsf{RS} \cap \mathrm{CFL}$ is a subset of the class of Clark's congruential CFLs."

**Response:** Added as Proposition 9.7. Clark (2010) places substitutable and $(k, \ell)$-substitutable languages in the congruential family; the revised text also explicitly identifies his Dyck example. The proposition extends the containment slice-wise to every fixed $s$, hence to $\mathsf{RS}(\Gamma) \cap \mathrm{CFL}$. Properness over $\{a, b\}$ is witnessed by $D_1$, which is congruential but lies outside $\mathsf{RS}$.

**Revision location:** Proposition 9.7.

## Reviewer 2

For the learning-algorithm correction, see R1.1 and R1.24; for the terminal-rule completeness repair raised below, see R2.8. We also recognize the reviewer's overall concern that the learning contribution and motivation for the fixed-typing setting were not sufficiently clear in the submitted version; R2.1 now separates inherited learning results from the new quantitative and expressive contributions.

#### General positioning of fixed-$h$

**R2.1. Comment (quoted excerpt):** "At a general level, the paper offers an abstraction of the learning of Yoshinaka's $(k, \ell)$-substitutable languages by replacing the fixed-length window by an arbitrary recognizable congruence, and lifts the known results accordingly. The abstraction places the problem in a powerful and well-understood framework, but it requires a whole morphism to be known in advance for each language, whereas the earlier settings fixed a single uniform parameter valid across the entire class. The paper acknowledges that learning the morphism is outside its scope, yet the interest of the setting – beyond providing a more abstract framework for the study of $(k, \ell)$-substitutable languages – remains unclear. A promising direction would be to deepen the links with pushdown automata, which Sections 8 to 10 already touch on. The links with the work of Takada ... are also worth investigating. In a more practical direction, this work could be read as extending automaton typing ... to context-free substitutable languages, and the conditions a typing must satisfy could be studied."

**Response:** We agree that supplying a finite-semigroup morphism is a stronger prior bias than fixing $(k, \ell)$, and we do not infer this morphism. However, $s$ is fixed uniformly across the entire target class $\mathcal{C}_s^{\mathrm{cf}}$; it is not chosen anew for each individual target language. For $s = s_{k,\ell}$, our reconstruction specializes to Yoshinaka's 2008 substring-rule construction, up to the stated representation choices.

The qualitative identification guarantees and polynomial-time hypothesis updates are specializations of Yoshinaka's 2015 general many-sorted framework, not independent novelties. Beyond the fixed-window encoded-size bound already obtained by Yoshinaka (2008, Theorem 1), the new content is quantitative and expressive. First, we obtain characteristic-data encoded-length bounds using ordinary thickness for window–finite-group typings (Corollary 7.5), and a grammar-size-only bound for linear targets (Theorem 8.3). Second, a fixed two-element semilattice typing yields finite targets with grammar size and ordinary thickness $O(n)$ but requiring

a characteristic-sample word of length $3 \cdot 2^n - 2$ for every set-driven operator admitting characteristic samples for the specified distinguishing pair (Proposition 6.4); Corollary 6.5 gives the corresponding intrinsic typed-thickness obstruction. This contrasts with examples whose ordinary thickness is already exponential. Third, nonregular linear and nonlinear examples lie outside every fixed-window class (Propositions 8.4 and 9.1), while the counter/Dyck results limit what fixed finite typing can represent (Section 9). These results do not provide a method to infer $s$, a uniform bound for all fixed typings, or a convergence-stage lower bound for arbitrary Gold learners.

We have also clarified the algebraic scope. Yoshinaka (2008, Proposition 1) established closure of the unrestricted fixed-$(k, \ell)$-substitutable class under intersection of two class members and under nonerasing inverse homomorphisms, as well as nonclosure under arbitrary regular intersection and arbitrary inverse homomorphisms; its context-free subclass need not be closed under intersection. Proposition 3.2 records elementary finite-typing analogues, including regular filtering and erasing inverse images after changing the typing; these closure facts are not claimed as new theorems. Section 3 also describes adequate typing as finite-index congruence separation of conflicting distributions, not a procedure for finding the typing. Finally, the revised Introduction cites automata typing bias (Coste et al., 2004), and Section 8 gives a limited connection with Takada's regular control sets for linear grammars. We claim neither a general pushdown characterization nor an equivalence with Takada's learning framework.

**Revision location:** Introduction; Section 3 (concise Yoshinaka comparison immediately before Proposition 3.2); Section 4; Propositions 3.2 and 3.5; Main Theorem 3.4(iv); Proposition 6.4, Corollary 6.5, and Appendix C; Section 7, Lemmas 7.4 and 7.6, Corollary 7.5, Appendix A.1; Proposition 8.4; Section 9.2 ($L_{\mathrm{Yreg}}$ definition); the control-set discussion in Section 8.

**Main concern 1: outer context typing and transport**

**R2.2. Comment (quoted excerpt):** "My main concern is the typed refinement, and specifically transport. Rule (2) allows a nonterminal to move between arbitrary contexts with no condition whatever on their types. Is this intended? If the outer components $m, n$ record where a nonterminal may appear, why does the rule that relocates it ignore them entirely? As far as I can tell the components are then read by no proof: Lemma 4.6 is invoked only for the yield component, and replacing $A_p^{m,n}$ by $A_p$ throughout leaves every result intact. Does the author agree, and if not, where exactly is an outer type consulted? The situation seems to me to follow from Definition 2.3 itself, whose algebraic condition bears on the substituted fragment and imposes nothing on the surrounding context – so that a learner mirroring that definition has nothing to guard transport with. If that is right, then the typed transport announced in Section 1.1 does not occur anywhere in the article, and the claim that occurrences must be separated according to their outer types cannot be established by any example. The yield typing, which is genuinely essential and is what becomes vacuous under the trivial monoid, should not be left in a footnote."

**Response:** We agree, and the observation permits a stronger simplification than merely deleting the outer type annotations. No proof in the *submitted version* consulted the outer components, so $A_p^{m,n}$ has been replaced by the yield-typed $A_p$ on the target side. On the hypothesis side, the context index is also inessential: submitted Rule (2) made all symbols $[x : u, v]$ with the same factor $x$ mutually derivable. Quotienting those symbols therefore gives one nonterminal $[\![x]\!]$ per observed factor. Submitted Rule (3) then becomes exactly the unary rule $[\![x]\!] \to [\![y]\!]$ when $x$ and $y$ share an observed context and $s(x) = s(y)$. The separate transport rule disappears. Accordingly, we withdraw the submitted claim that occurrences must be separated by their outer types, as well as the typed-transport claim in submitted Section 1.1.

Apart from the start-symbol presentation, the treatment of $\lambda$, and redundant identity rules, the revised batch grammar follows Yoshinaka's substring-indexed rule scheme (2008, Section 4), also presented by Clark (2013, Section 4). For window typing $s_{k,\ell}$ its nontrivial unary rules coincide with Yoshinaka's; for general fixed $s$ they require type equality and a shared sample context. Soundness rests on the invariant of Theorem 4.2, $[\![x]\!] \Rightarrow^* w$ implies $x \equiv_L w$ and $s(w) = s(x)$ (R1.27), which consults no outer type.

Yield typing, which the reviewer rightly identifies as essential, is now discussed in the main text of Section 5.1 rather than in a footnote. It is needed only for the present witness-based completeness argument, where it licenses the unary steps from a canonical parent yield to a typed terminal yield or to a canonical right-hand-side yield. This is a requirement of the witness-based completeness proof, not a claim of algorithm-independent necessity.

**Revision location:** Section 4; Section 5.1 and Theorem 5.2.

**Main concern 2: extent of the linear theorem**

**R2.3. Comment (quoted excerpt):** "The linear result raises a different question: how large is the class it covers? The three closing sections situate RS, not $\mathcal{C}_h^{\mathrm{lin}}$, so the reader is given no sense of what Theorem

7.9 reaches beyond the linear substitutable languages. The capped-counter languages are regular and hence linear, so the separation from Yoshinaka's hierarchy already holds inside a linear class, whereas $\{a^n b^n\}^*$ is not linear, so the final section establishes nothing at that level. As it stands the linear theorem reads as a technical strengthening on a class of unknown extent, the non-regular linear case being left open."

**Response:** Section 8 now provides both a concrete nonregular linear example and a structural closure mechanism. Proposition 8.4 proves that $L_{\pm,e}$ is linear, nonregular, and substitutable under a fixed finite-semigroup typing, but belongs neither to the Clark–Eyraud substitutable class nor to any $(k,\ell)$-substitutable class. More generally, Proposition 3.2(ii), together with closure of linear languages under regular intersection, gives the implication used in the proof of Proposition 8.4:

$$L \in \mathcal{C}_s^{\mathrm{lin}}, \quad R_{\mathrm{reg}} = g^{-1}(\mathsf{Acc}_g) \quad \Longrightarrow \quad L \cap R_{\mathrm{reg}} \in \mathcal{C}_{s\times(g|_{\Sigma^+})}^{\mathrm{lin}}.$$

Here $t = g|_{\Sigma^+}$, with codomain $g(\Sigma^+)$, as in Proposition 3.2(ii). For a suitable acceptance set $\mathsf{Acc}_t \subseteq g(\Sigma^+)^\sharp$, $g^{-1}(\mathsf{Acc}_g) = \widehat{t}^{-1}(\mathsf{Acc}_t)$: the fresh identity records whether $\lambda$ is accepted. That proof writes $L_{\pm,e} = L_{\mathrm{all}} \cap R_{\mathrm{reg}}$ with $L_{\mathrm{all}} = \{a^n \xi b^n : n \geq 0,\ \xi \in \{c,d,e\}\}$, which is linear and Clark–Eyraud substitutable under the nonempty-factor convention, and $R_{\mathrm{reg}} = (aa)^* c (bb)^* \cup a(aa)^* d\, b(bb)^* \cup a^* e b^*$; a subsequent paragraph exhibits a more economical four-element typing with a short verification. The regular-level separation was already known from Yoshinaka (2008, Proposition 1); the capped-counter family is retained only as the finite-state companion to the uncapped counter CTR. No separation from Takada's control-set approach is claimed.

We withdraw the submitted Section 7 title, *The Linear Subclass: A Full Polynomial Time-and-Data Theorem*, as a characterization of the sequential learner. Theorem 8.3 now establishes polynomial characteristic data only for the set-driven operator $\mathsf{Rec}_s$, relative to a chosen linear CFG representation of a fixed-$s$ substitutable target, with no incremental polynomial-data guarantee for $\mathsf{Learn}_s$. This does not contradict de la Higuera's negative result for the unrestricted class of linear languages: our bound applies only to $\mathcal{C}_s^{\mathrm{lin}}$ for fixed $s$.

**Revision location:** Proposition 3.2(ii); Theorem 8.3 and its scope clarification; Proposition 8.4 (regular-filter argument in its proof) and the following four-element typing paragraph.

**Further comments**

**R2.4. Comment (quoted excerpt):** "The paper would gain considerably from being simplified: much of Section 4 is definitional, several results restate what their proofs have just established, and objects are regularly used before they are introduced, which forces the reader to hold material in suspense. Relatedly, since $\widetilde{G}$ is by construction the trimmed refinement of $\widetilde{G}_{\mathrm{full}}$, its symbols are reachable and productive by definition; repeating this qualification at every occurrence adds length without adding content, and one sentence recording the convention would suffice."

**Response:** We agree. The repeated reachable/productive qualifications have been removed and the reduced-refinement convention is stated once in Section 5.1. Redundant results, including former Lemmas 4.6–4.7 and 5.3–5.4 and Theorems 4.9 and 4.12, are deleted (R1.22, R2.9). To avoid forward use, the batch reconstruction operator is defined before the target-side refinement, $\omega$ before its first use, $\chi$ immediately before the witness set, and *locally trivial* in Section 2.2 before Proposition 3.5. The quantitative material is separated into Sections 6 (general typed thickness), 7 (window–finite-group typings), and 8 (linear targets), with long normalization proofs in the appendices.

**Revision location:** Section 4; Section 5.1; Sections 6–8; Appendices A–B.

**R2.5. Comment (quoted excerpt):** "Proposition 3.1: the hypothesis $L = h^{-1}(F)$ characterises the regular languages exactly. Replace 'In particular' by 'Thus' (and the proof could make explicit why)."

**Response:** Proposition 3.1 states the standard regular-language recognition characterization in the finite-semigroup formulation, using the canonical fresh-identity extension $\widehat{s}$ to distinguish $\lambda$. If $s(x) = s(y)$ for positive $x, y$, then $\widehat{s}(uxv) = \widehat{s}(uyv)$ for every outer context. Thus any language recognized by $\widehat{s}$ is $\sim_s$-substitutable, establishing the required consequence. We cite Eilenberg (1974, Chapter III) for the classical result.

**Revision location:** Proposition 3.1.

**R2.6. Comment (quoted excerpt):** "Definition 4.11: the basis $B(\widetilde{G})$ is never used. Is it useful to introduce it as well as Theorem 4.12 and Remark 4.13? (Notations NT() and Rule() are unnecessary since it is sufficient to define $G = ()$ and use defined $V$ and $P$.)"

**Response:** Agreed. These objects and notations have been removed.

**Revision location:** Section 5.1.

**R2.7. Comment (quoted excerpt):** "Example 4.3 is not in SSBNF"

**Response:** The former example has been removed entirely; the revised proof no longer needs it.

**Revision location:** Section 5.1.

**R2.8. Comment (quoted excerpt):** "Lemma 5.2 (i): If $X$ carries several terminal rules $\omega(X)$ is not sufficient, and rule (3) should be used (proof needs to be corrected accordingly)."

**Response:** The reviewer identified a genuine terminal-rule completeness gap. The revised Section 5 uses an arbitrary typed CFG production $\rho : X \to u_0 Y_1 u_1 \cdots Y_r u_r$. For each retained typed rule, the witness set contains $u_X t_\rho v_X$, where $t_\rho$ replaces each typed nonterminal child by its canonical yield. In particular, a terminal rule $X \to a$ yields the witness $u_X a v_X$. Together with $u_X \omega(X) v_X$, this supplies the shared context and equal positive $s$-type needed for unary Rule (U); repeated Rules (B) and (L) then emit any terminal-only right-hand side. For general right-hand sides, Rule (B) splits the nonempty components and induction handles the positive-type children; children of fresh-identity type contribute only $\lambda$. Thus unit, nullable, and mixed productions are handled without imposing BUNF in the completeness proof (Theorem 5.2).

**Revision location:** Section 5.1 (witness set); Theorem 5.2 (terminal-rule base case).

**R2.9. Comment (quoted excerpt):** "Are Lemmas 5.3 and 5.4 needed?"

**Response:** Agreed. They are deleted.

**Revision location:** Sections 4–5.1.

**R2.10. Comment (quoted excerpt):** "Corollary 5.8 Should begin with 'For a fixed homomorphism $h$ from $\Sigma^*$ to a finite monoid $M$, ....', or 'Being given a homomorphism $h$ from....' "

**Response:** Corollary 5.4 now begins with the analogous more general assumption: *For a fixed surjective semigroup homomorphism* $s : \Sigma^+ \to \mathsf{S}$ *to a finite semigroup.* As requested, the homomorphism is fixed uniformly across its target class.

**Revision location:** Corollary 5.4.

**R2.11. Comment (quoted excerpt):** "section 7.1 Transformation to standard SSLNF (3 pages) could be condensed or given in appendix."

**Response:** We took the appendix option: Section 8 retains only Proposition 8.1 and Lemma 8.2, used directly by Theorem 8.3, with the short proof of Lemma 8.2; the normalization proof is in Appendix B.

**Revision location:** Proposition 8.1 and the surrounding discussion; Lemma 8.2; Theorem 8.3 and the following clarification; Appendix B.